WORKED SOLUTION

STEP I 1996, Question 6

Pure • Trigonometric Sums and Integrals

STEP I 1996, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q6

The key idea

The given identity is built to telescope: if we multiply the sum $\sum \cos kx$ by $2\sin\frac{1}{2}x$, each term becomes a difference of two sines, and neighbouring differences cancel. Once f is written as a sum of cosines, the awkward-looking integrals in (ii) become integrals of single cosines, most of which vanish over $[0, \pi]$.

Solution

Throughout, n is a positive integer.

Part (i)

Put $S = \sum_{k=1}^{n} \cos kx$. Multiplying by $2\sin\frac{1}{2}x$ and using the given formula term by term,

$$\begin{aligned} 2\sin\tfrac{1}{2}x\, S &= \sum_{k=1}^{n} \Big(\sin(k+\tfrac{1}{2})x - \sin(k-\tfrac{1}{2})x\Big) \\ &= (\sin\tfrac{3}{2}x - \sin\tfrac{1}{2}x) + (\sin\tfrac{5}{2}x - \sin\tfrac{3}{2}x) + \cdots + (\sin(n+\tfrac{1}{2})x - \sin(n-\tfrac{1}{2})x) \\ &= \sin(n+\tfrac{1}{2})x - \sin\tfrac{1}{2}x, \end{aligned}$$

since every intermediate term appears once with each sign and cancels. Hence

$$\sin(n+\tfrac{1}{2})x = \sin\tfrac{1}{2}x + 2\sin\tfrac{1}{2}x \sum_{k=1}^{n} \cos kx.$$

For $0 < x \leq \pi$ we have $0 < \frac{1}{2}x \leq \frac{1}{2}\pi$, so $\sin\frac{1}{2}x > 0$ and we may divide by it:

$$\mathrm{f}(x) = \frac{\sin(n+\frac{1}{2})x}{\sin\frac{1}{2}x} = 1 + 2\sum_{k=1}^{n} \cos kx,$$

as required.

Part (ii)

Ask yourself f isn't defined at $x = 0$. Is $\int_0^\pi f(x)\,dx$ even a sensible integral?

Yes. By part (i), on $0 < x \leq \pi$ the function f agrees with the cosine polynomial $1 + 2\sum \cos kx$, which is continuous on all of $[0, \pi]$ (with value $2n + 1$ at 0). Changing or leaving undefined the value at a single point doesn't affect an integral, so we may integrate the cosine sum instead.

First integral. For each integer $k \geq 1$,

$$\int_0^\pi \cos kx\,dx = \left[\frac{\sin kx}{k}\right]_0^\pi = 0,$$

because $\sin k\pi = 0$. So only the constant term survives:

$$\int_0^\pi f(x)\,dx = \int_0^\pi 1\,dx + 2\sum_{k=1}^{n} \int_0^\pi \cos kx\,dx = \pi.$$

Second integral. Now

$$\int_0^\pi f(x)\cos x\,dx = \int_0^\pi \cos x\,dx + 2\sum_{k=1}^{n} \int_0^\pi \cos kx \cos x\,dx.$$

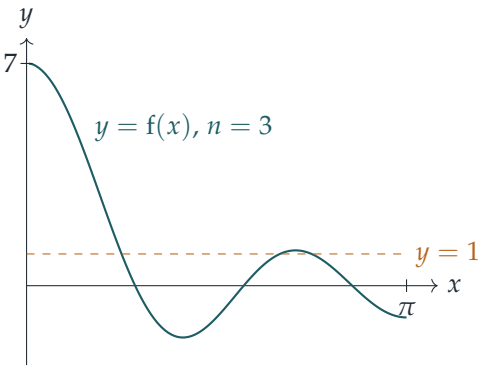
The first integral is $[\sin x]_0^\pi = 0$. For the others, use $2\cos kx \cos x = \cos(k+1)x + \cos(k-1)x$:

$$2\int_0^\pi \cos kx \cos x\,dx = \int_0^\pi \cos(k+1)x\,dx + \int_0^\pi \cos(k-1)x\,dx.$$

Here $k + 1 \geq 2$, so the first integral is 0 as above. The second is also 0 unless $k - 1 = 0$, when the integrand is 1 and the integral is π. So only the $k = 1$ term contributes, and

$$\int_0^\pi f(x)\cos x\,dx = \pi.$$

(If n were allowed to be 0, then $f \equiv 1$ and this integral would be 0; for every $n \geq 1$ it is π.)



The picture shows why the first answer doesn't depend on n: the tall spike near 0 (height $2n+1$) is balanced by the oscillations, and the net area is always that of the constant term, π.

Checks

- As $x \to 0$, $\sin(n+\frac{1}{2})x / \sin\frac{1}{2}x \to (2n+1)$, and the cosine sum at $x=0$ is $1+2n$. They agree.
- At $x=\pi$: $\mathrm{f}(\pi) = \sin(n+\frac{1}{2})\pi = (-1)^n$, and $1 + 2\sum_{k=1}^{n}(-1)^k = (-1)^n$ too.
- Midpoint-rule quadrature of the original quotient for $n = 1,\ldots,7$ gives $3.14159265\ldots$ for both integrals, matching π.

Common slips

- Forgetting to justify dividing by $\sin\frac{1}{2}x$: it's nonzero precisely because $0 < x \leq \pi$.
- Writing $\int_0^\pi \cos kx \cos x \,\mathrm{d}x = 0$ for all k, and so getting 0 for the second integral; the $k=1$ term gives $\int_0^\pi \cos^2 x \,\mathrm{d}x = \frac{1}{2}\pi$.

Beyond STEP

The function f is the *Dirichlet kernel*, the central object in the theory of Fourier series. The calculations in (ii) are instances of orthogonality: $\cos kx$ and $\cos mx$ integrate to zero against each other over $[0,\pi]$ unless $k=m$. That's exactly how Fourier coefficients are extracted, and integrating a function against the Dirichlet kernel gives the nth partial sum of its Fourier series. For a careful treatment of when those partial sums converge, see Abbott, *Understanding Analysis*.

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